

FORT MACLEOD AGCM, AB, Canada

WMO: 717890

Lat: 49.7858N

Lon: 113.3794W

Elev: 3209

StdP: 13.07

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.2	-13.8	-25.8	1.5	-19.9	-19.2	2.2	-13.7	41.2	37.5	36.8	38.5	5.4	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	87.0	60.3	83.4	59.5	80.2	58.8	64.7	77.6	62.7	77.2	60.9	75.3	11.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	86.8	70.9	57.6	79.8	68.5	55.5	73.8	65.3	31.5	77.2	29.9	77.2	28.6	75.4	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.6	29.7	25.9	-31.0	93.7	5.5	4.4	-34.9	96.9	-38.1	99.5	-41.2	101.9	-45.2	105.1		
			-31.2	69.2	5.4	2.2	-35.1	70.8	-38.3	72.1	-41.3	73.3	-45.3	74.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	23.0	20.7	30.9	40.4	50.4	57.7	63.6	62.8	54.7	42.5	31.1	19.7	(6)
(7)		DBStd	19.28	15.27	16.11	13.77	9.04	7.67	5.68	4.88	5.83	8.31	10.24	13.90	16.30	(7)
(8)		HDD50	4476	839	819	594	302	89	6	0	2	46	269	572	938	(8)
(9)		HDD65	8648	1303	1239	1057	738	453	228	85	110	317	699	1016	1403	(9)
(10)		CDD50	1407	1	0	2	14	103	238	421	398	188	36	6	0	(10)
(11)		CDD65	103	0	0	0	0	2	9	41	42	9	0	0	0	(11)
(12)		CDH74	2590	0	0	0	14	133	241	910	944	341	7	0	0	(12)
(13)		CDH80	753	0	0	0	1	19	52	270	320	91	0	0	0	(13)
(14)	Wind	WSAvg	10.8	12.2	10.7	11.2	11.3	9.8	10.4	9.1	8.7	9.9	11.9	12.6	12.3	(14)
(15)	Precipitation	PrecAvg	16.7	0.7	0.6	1.0	1.5	2.2	3.0	1.7	1.8	1.7	0.7	0.7	0.8	(15)
(16)		PrecMax	24.1	1.9	1.3	3.0	4.8	7.2	8.5	5.4	4.8	5.7	2.1	2.1	3.2	(16)
(17)		PrecMin	10.9	0.1	0.0	0.2	0.1	0.9	0.3	0.2	0.0	0.2	0.0	0.0	0.1	(17)
(18)		PrecStd	3.3	0.5	0.4	0.6	1.2	1.3	2.0	1.4	1.3	1.3	0.6	0.6	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	55.7	61.5	75.8	82.7	87.4	91.4	92.4	87.2	73.5	64.6	51.6	(19)
(20)		MCWB	40.2	42.1	44.8	51.3	55.2	62.4	60.9	60.2	58.0	52.7	47.5	41.2		(20)
(21)		2%	DB	48.9	50.1	56.9	67.6	77.9	79.9	86.4	87.3	82.5	67.8	57.7	47.5	(21)
(22)		MCWB	39.3	39.1	42.9	47.7	54.6	58.7	61.0	60.1	57.2	49.9	45.5	38.4		(22)
(23)		5%	DB	45.3	45.9	53.3	62.0	72.7	76.1	82.7	83.4	78.0	62.8	52.8	43.7	(23)
(24)		MCWB	37.0	36.4	41.2	45.2	52.8	57.7	60.7	59.3	55.8	48.4	42.6	36.2		(24)
(25)		10%	DB	42.4	42.1	49.1	57.1	68.0	72.5	79.3	79.4	72.3	58.1	48.6	40.8	(25)
(26)		MCWB	35.4	34.5	38.8	42.8	50.7	55.9	60.6	58.5	54.1	45.5	39.9	34.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	43.4	46.4	52.1	57.6	65.3	69.1	65.0	60.9	55.1	49.2	42.5	(27)
(28)		MCDWB	52.5	53.6	58.1	73.4	76.4	78.7	80.8	79.0	78.6	69.8	62.0	50.3		(28)
(29)		2%	WB	39.6	40.0	44.0	48.7	55.4	62.0	65.8	63.0	58.6	51.9	46.2	39.1	(29)
(30)		MCDWB	48.0	49.4	55.3	64.8	75.1	76.5	78.1	79.7	77.7	65.7	57.6	45.8		(30)
(31)		5%	WB	37.7	37.0	41.4	45.9	53.6	59.4	63.9	61.5	56.7	48.9	43.0	36.8	(31)
(32)		MCDWB	44.6	45.0	52.0	59.9	71.2	72.3	76.2	77.8	74.2	60.6	52.0	43.1		(32)
(33)		10%	WB	35.8	34.6	39.1	43.4	51.7	57.1	61.9	59.9	54.7	46.3	40.1	34.4	(33)
(34)		MCDWB	42.3	41.8	48.1	55.8	66.0	69.6	75.0	75.7	70.9	57.4	48.2	40.9		(34)
(35)	Mean Daily Temperature Range	MDBR	19.1	20.9	20.9	23.4	24.1	23.7	27.3	28.5	26.3	22.5	19.6	18.9		(35)
(36)		5% DB	MCDBR	18.8	23.7	26.1	33.5	34.3	31.9	35.5	37.5	37.6	28.7	22.2	19.1	(36)
(37)		MCWBR	12.1	14.7	14.6	17.0	14.8	14.5	15.1	14.2	15.7	14.6	12.8	13.1		(37)
(38)		5% WB	MCDBR	19.1	23.5	24.8	31.9	31.2	28.7	29.1	31.3	34.0	27.3	22.9	19.1	(38)
(39)		MCWBR	13.3	15.5	14.6	16.5	13.8	14.1	14.9	13.7	14.7	14.3	13.7	13.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.259	0.281	0.319	0.330	0.327	0.332	0.346	0.310	0.282	0.258	0.232		(40)
(41)		taud	2.435	2.467	2.479	2.388	2.417	2.471	2.468	2.410	2.484	2.537	2.499	2.448		(41)
(42)		Ebn at Noon	262	289	298	294	294	295	292	282	283	275	258	256		(42)
(43)		Edh at Noon	19	23	28	34	35	33	33	33	28	22	18	16		(43)
(44)	All-Sky Solar Radiation	RadAvg	392	709	1112	1535	1830	2031	2230	1774	1259	783	436	308		(44)
(45)		RadStd	32	57	91	104	149	145	115	110	107	81	36	29		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (67 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page